

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 37

BIODIVERSITY - 3

BIODIVERSITY

BIODIVERSITY HOTSPOTS OF INDIA

2 THE WESTERN GHATS

- The Western Ghats is a major biodiversity area. It covers about 4.0 per cent of the total geographical area.
- It is one of the major tropical evergreen forest regions in India.
- The zone stretches from the hills to the South of the Tapti River in the North to Kanyakumari in the South.

	TOTAL SPECIES	ENDEMIC SPECIES
Plants	5916	3049
Mammals	140	18
Birds	458	35
Reptiles	267	174
Amphibians	178	130
Freshwater Fishes	191	139



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BIODIVERSITY HOTSPOTS OF INDIA

2 THE WESTERN GHATS

- Significant species endemic to this region include
 - ✓ Nilgiri Langur
 - ✓ Lion Tailed Macaque
 - ✓ Grizzled Giant Squirrel
 - ✓ Malabar Civet
 - ✓ Nilgiri Tahr
 - ✓ Malabar Grey Hornbill
 - ✓ The Travancore Tortoise and Cane turtle are two species restricted to a small area in central Western Ghats.



NILGIRI LANGUR



LION TAILED MACAQUE



MALABAR CIVET

2 THE WESTERN GHATS

- The Western Ghats intercept the southwestern monsoon winds. The western slopes of the mountains experience heavy annual rainfall, while the eastern slopes are drier.
- Rainfall also decreases from south to north.
- The wide variation of rainfall patterns in the Western Ghats, coupled with the region's complex geography, produces a great variety of vegetation types.
- These include scrub forests in the low-lying rain shadow areas and the plains, deciduous and tropical rainforests up to about 1,500 m, and a unique combination of montane forests and rolling grasslands (shola) above 1,500 m.



2 THE WESTERN GHATS

- The Western Ghats contain the **most extensive tropical forests** in the Indian Peninsula in the States of Kerala, Karnataka and Tamil Nadu.
- **Geological stability over a long period of time, proximity to the equator and high rainfall** have contributed to the tropical moist forests in the Western Ghats being endowed with **exceptional ecological & species diversity and endemism.**
- Forests south of the Palakkad Gap along the Kerala - Tamil Nadu boundary have the **richest biodiversity** and higher endemism than the rest of the Western Ghats.
- The Western Ghats also harbor a large diversity of human cultures. For example, in the less than 20,000 sq.km. of Kerala Western Ghats there are more than **38 distinct tribal communities**



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2 THE WESTERN GHATS

ISSUES IN WESTERN GHATS

- The biggest ecological damage inflicted upon the Western Ghats is deforestation. Within the past century more than 50 percent of the forest vegetation has been lost.
- The remaining forests are extremely fragmented and most of the fragments are not viable ecologically.
- Climax plant communities occupy only a tiny fraction of the area, less than 10 percent, and even these are getting rapidly degraded.
- Later tree plantations (mostly teak and eucalyptus) replaced practically all the lower elevation forests. Converting natural forests into monoculture plantations, essentially as industrial raw material sources.
- Such plantations occupy about 30 percent of the total forest cover.



BIODIVERSITY HOTSPOTS OF INDIA

3 INDO-BURMA REGION

- The hotspot begins in eastern Bangladesh and then extends across north-eastern India, south of the Brahmaputra River, covers all of Myanmar, part of southern China, all of Laos, Cambodia and Vietnam, Thailand and a small part of Peninsular Malaysia.
- It also includes the Andaman Islands.



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BIODIVERSITY HOTSPOTS OF INDIA

3 INDO-BURMA REGION

WHY IS IT A BIODIVERSITY HOTSPOT?

- Covering more than 2 million km² of tropical Asia, Indo-Burma is still revealing its biological treasures. Six large mammal species have been discovered in the 90s and 2000s.
- This hotspot also holds remarkable endemism in freshwater turtle species, most of which are threatened with extinction, due to extensive habitat loss.
- Bird life in Indo-Burma is also incredibly diverse, holding almost 1,300 different bird species.
- An estimate of total plant diversity in the hotspot reveals about 13,500 species, of which about 7,000 (52%) are endemic.

	TOTAL SPECIES	ENDEMIC SPECIES
Plants	13,500	7000
Mammals	433	73
Birds	1266	64
Reptiles	522	204
Amphibians	286	154
Freshwater Fishes	1262	553



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3 INDO-BURMA REGION

ISSUES IN INDO-BURMA REGION

- With over 500 million people, Indo-Burma has the largest human population of the world's 36 hotspots.
- This is reflected in the fact that remaining natural habitat is restricted to only five percent of its original extent.
- The large and rapidly expanding human population, coupled with some of the fastest rates of economic growth in the world, is placing increasing pressures on remaining natural ecosystems.
- Expansion of agriculture (such as rice, rubber and oil palm), infrastructure development (especially roads and hydropower dams), timber extraction and illegal trade in wildlife are the major current threats to the hotspot's biodiversity.



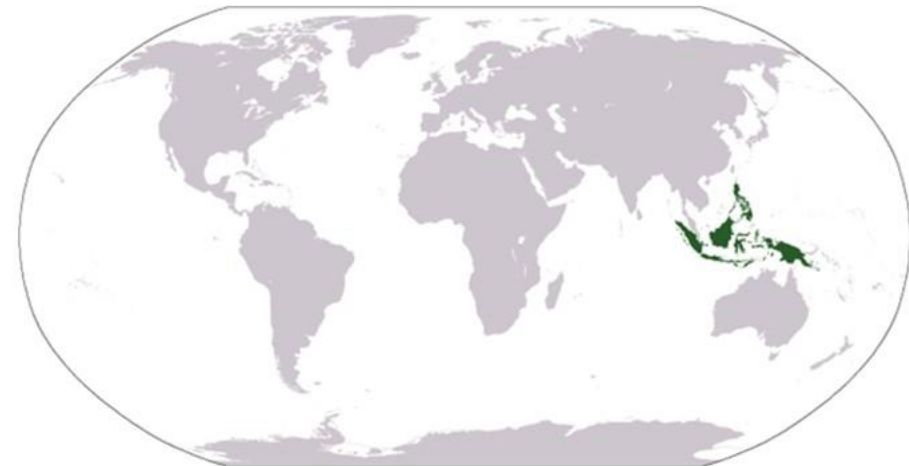
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BIODIVERSITY HOTSPOTS OF INDIA

4 SUNDALANDS

- Sundaland covers a small portion of southern Thailand, nearly all of Malaysia, Singapore at the tip of the Malay Peninsula, all of Brunei and all of the western half of Indonesia, The Nicobar Islands are also included.
- The Sundaland hotspot covers the western half of the Indo-Malayan archipelago, an arc of 17,000 equatorial islands, which include two of the largest islands in the world: Borneo and Sumatra.

	TOTAL SPECIES	ENDEMIC SPECIES
Plants	25000	15000
Mammals	380	172
Birds	769	142
Reptiles	452	243
Amphibians	244	196
Freshwater Fishes	950	350



4

SUNDALANDS

THE GREAT NICOBAR BIOSPHERE RESERVE

- The Great Nicobar Biosphere Reserve covers a large part (85%) of the island of Great Nicobar, the largest of the Nicobar Islands.
- The Reserve was declared in 2013. It incorporates two National parks: the larger Campbell Bay National Park on the northern part of the island, and Galathea National Park in the southern interior.



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BIODIVERSITY HOTSPOTS OF INDIA

4

SUNDALANDS

THE GREAT NICOBAR BIOSPHERE RESERVE

- The non-Biosphere portions of the island (set aside for agriculture, forestry and settlements) are confined to the southwestern and southeastern coastal reaches.
- A total of 14 species of mammals, 71 species of birds, 26 species of reptiles, 10 species of amphibians and 113 species of fish have been reported. Of which, 11 species of mammals, 32 species of birds, 7 species of reptiles and 4 species of amphibians have been found to be endemic.
- Crab-eating Macaque, Nicobar Tree Shrew, Dugong, Nicobar Megapode, Serpent Eagle, Salt-water Crocodile and Reticulated Python are endangered.



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